Michael Schmalz

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Education

Iowa State University - Ames, Iowa Expected Graduation: May 2027

Bachelor of Science in Computer Engineering

Experience

Vermeer Corporation - Pella, Iowa May 2026 - Aug 2026

Embedded Software Engineer Intern

- Developed and modified embedded machine software using MATLAB and Simulink to support software updates and machine functionality.
- Utilized J1939 and UART communication protocols to support communication between embedded systems and machine components.
- Built an application to streamline software updates for production floor personnel, technical support teams, and engineers.
- Collaborated with technical teams in an Agile environment to troubleshoot issues, coordinate development efforts, and deliver solutions efficiently.
- Quickly learned new software tools, technologies, and development workflows to adapt to project requirements and contribute to ongoing engineering tasks.

T&M Services - Des Moines, Iowa May 2025 - Aug 2025

Installer

- Partnered in teams of 2-5 to install and relocate a variety of office furniture systems, including heavy and fragile items, across 15,000+ sq. ft. commercial spaces, ensuring efficiency and safety.
- Provided seamless, move-in-ready environments for clients by exceeding their highest standards.
- Operated 26-ft trucks to transport furniture, equipment, and site waste between job locations, guaranteeing clean and organized job sites.

Hy-Vee Supermarket - West Des Moines, Iowa Sept 2021 - Aug 2024

Stocker and Checker

- Unloaded 2-3 incoming shipments per day and maintained a streamlined and functional backroom, improving inventory accessibility and workflow efficiency.
- Assisted dozens of shoppers per shift concerning product location and various questions, creating an effortless shopping experience.

Projects

Autonomous Vacuum Robot - Embedded Systems Jan 2025 - May 2025

- Developed an autonomous vacuum robot in embedded C on the Tiva TM4C123GH6PM microcontroller, integrating IR, ultrasonic (PING), bumper, and cliff sensors.
- Navigated to a goal destination via Wi-Fi around 5 different obstacle types in a 30 sq. ft. area using onboard sensing and control logic.

RISC-V Processor - Computer Organization and Assembly Level Programming Aug 2025 - Dec 2025

- Designed, implemented, and verified a 32-bit RISC-V RV32I processor in VHDL, including single-cycle and both software-scheduled and hardware-scheduled 5-stage pipelined architectures.
- Performed hazard detection, data forwarding, and pipeline stalling, achieving 2x cycle-time speed up.
- Built comprehensive assembly tests verifying the correct execution of 35 unique instructions and real programs totaling 2,000+ instructions, simulating real-world verification workflows.

Activities and Leadership

- CyMATH Tutor Nov 2025 - Present
  - Mentored high school students in mathematics and introductory engineering concepts.
- Institute of Electrical and Electronics Engineers at Iowa State Aug 2024 - Present
- Pickleball Club Aug 2024 - Present

Technical Skills

Languages: C, C++, Python, Java, JavaScript, VHDL, RISC-V Assembly, SQL

Software: GitHub, Linux, QuestaSim, Quartus II, LTspice, Eclipse, Visual Studio Code, Simulink